

Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

Annamaria Iannuzzo

Delft University of Technology

Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

by

Annamaria Iannuzzo

to obtain the degree of Master of Science at the Delft University of Technology,
to be defended publicly on Monday January 1, 2024 at 10:00 AM.

Student number: 6202144
Project duration: March 25, 2023 – CHANGE DATE, 2024
Thesis committee: Prof. dr. ir. J. Doe, TU Delft, supervisor
Dr. E. L. Brown, TU Delft
Ir. A. Aaronson, Acme Corporation

*This thesis is confidential and cannot be made public until **CHANGE DATE** December 31, 2024.*

An electronic version of this thesis is available at <http://repository.tudelft.nl/>.

Cover: Canadarm 2 Robotic Arm Grapples SpaceX Dragon by NASA
under CC BY-NC 2.0 (Modified)



Preface

A preface...

*Annamaria Iannuzzo
Delft, October 2026*

Summary

A summary...

Contents

Preface	i
Summary	ii
Nomenclature	vii
1 Introduction	1
1.1 Motivation	1
1.2 Project framing	2
1.2.1 Care centralisation and territorial inequity in the Italian healthcare system	2
1.2.2 Prior research and positioning of the present thesis	2
1.2.3 Stakeholders	3
1.3 Document Outline	3
2 Background and Research-Development Gap	5
2.1 Regional Market Analysis: medical UAV logistics in Piedmont	5
2.2 Airframe-first vs. Safety-first: a design-philosophy gap	8
2.3 Airbag impact attenuation: background and design variables	8
2.3.1 Established applications and common working principle	9
2.3.2 Design guidelines and variables	9
2.4 Research-Development Objective	10
3 Methodology - Airbag Design Workflow	11
3.1 Problem statement	11
3.1.1 Reference mission and design inputs	11
3.1.2 Information gap	13
3.2 Proposed approach	13
3.2.1 Outer design loop: formalising the <i>Safety-First</i> principle	13
3.2.2 Inner design loop: a payload-protection subsystem design workflow	15
4 Results - Through the Workflow	17
4.1 Mission and failure scenario analysis: calculating V_x and V_y	17
4.2 Studying the payload: calculating allowable impact acceleration	17
4.2.1 Material characterisation	17
4.2.2 Sled test	17
4.3 Design variables optimisation: the code!!!	17
4.4 Summary of results	18
5 Conclusions	19
5.1 General summary	19
5.2 Recommendations for future work	19
6 Project Management	20
6.1 Strategy and planning	20
6.2 Challenges	20
6.2.1 challenge 1: aligning interests for thesis scope	20
6.2.2 challenge 2: Computational resources NO MORE	20
6.2.3 challenge 3: IT infrastructure bottleneck	20
6.2.4 challenge 4: going against the judgement of supervisors	20
6.3 Lessons learned	20
References	21

A	Extended Market Analysis	25
A.1	Value-chain actors	25
A.2	Trials and planned initiatives	28
B	FEM Iterations for glass phial model	30
C	Task Division Example	31
	Acknowledgements - <i>Ringraziamenti</i>	32

List of Figures

3.1	Akineton Fiale packaging and single phial.	13
3.2	Sequence of design events in the two approaches: the order of the mission and safety loops is swapped.	14
3.3	Inner design loop, specific for the airbag-based payload-protection subsystem.	15

List of Tables

2.1	Actors involved in drone delivery of medical goods in Piedmont, grouped by value-chain stage.	6
2.2	Airbag design variables by design phase.	10
A.1	Actors involved in drone delivery of medical goods in Italy	25
A.2	Drone delivery of medical goods in Italy: trials and planned initiatives	28
C.1	Distribution of the workload	31

Nomenclature

If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m ³]
...		

Introduction

1.1. Motivation

The present MSc Thesis in Aerospace Structures and Materials falls under the category of *Engineering Design*, in alternative to the *Scientific Research*-type projects. In *design* projects, it is required to attempt the process of innovation not through the discovery and exploration of a research niche, but rather by applying pre-existing scientific knowledge to enable the concrete realisation of a technical design. In the case of the present thesis, the overall objective consists in proposing a *design workflow* and validating it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, in line with the requirements set by the project commissioners. Therefore, the successful completion of this thesis is achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the device's design.

The design workflow (informally called *inner (design) loop* throughout this thesis) is intended as a first attempt to bridge the gap between the long-term, broadly-defined objective of constructing a large-scale drone network (also called *outer (design) loop*) over the region of Piedmont, Italy, and the previously completed research work, consisting in a series of logistic, legislative, and generally high-level evaluations of possible **drone-based missions** with the objective to **deliver medical supplies in emergency scenarios**. Such work was conducted *Politecnico di Torino*, in collaboration with the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and automotive consulting company *FEV Italia S.r.l.*, three of the main industrial and research stakeholders working towards helping hospitals in the transition from road to air transport to enhance the logistics of the *Sistema Sanitario Nazionale (SSN)*, the Italian National Healthcare System) both at an intra- and inter-hospital level.

This MSc Thesis Project is, therefore, a multidisciplinary project where the knowledge and techniques of Structural Engineering are put to the service of the medical field. The airbag-like device resulting from the design process should be able to protect the free-falling medical payload in the event of a drone failure and its subsequent detachment from the latter. The device aims to bring the impact accelerations within the allowable limits of the payload container and of the medical payload being transported, in compliance with the safety regulations regarding dispersion of chemical substances into the environment. However, contrary to most aeronautical projects which do not distinguish between the integrity of the aerial vehicle and the safety of its passengers, this project adopts a "payload-first" approach where the protection of the highly valuable medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety aspects of the unmanned aerial vehicle (UAV) and its payload implies a certain facilitation in the certification process of the UAV, as well as for the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery

missions.

1.2. Project framing

1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transforming healthcare delivery networks from road- to air-transport-based systems stems from the consensus within the Italian medical field, consolidated during the public health emergency of the COVID-19 pandemic (2020–2022), that the current ambulance-based architecture of the SSN's logistic infrastructure (described in detail¹ in [1]) cannot accommodate the healthcare system's needs and transitions. The two main - and competing - factors are:

1. **Care centralisation:** [2] describes care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Unhomogeneous distribution of emergency services :** [3] affirms that, in contrast with the centralisation maneuver described above, emergency resources are still unhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [4] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the SSN, the logistics of the Italian healthcare supply chain.

1.2.2. Prior research and positioning of the present thesis

In response to the issues highlighted in Section 1.2.1, recent years have seen an increasing number of technology development initiatives where industry players in the transport sector and research institutions work together to push for the modernisation of transportation means within the Italian SSN. Several of these industry-research partnerships have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [5], ordinary logistics vs. emergency missions and humanitarian applications[6], experimental/direct flight between strategic poles [7] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study[8]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Torino), a research group within the Department of Mechanical and Aerospace Engineering at *Politecnico di Torino*, and the Numerical Analysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company *FEV*, a global leader in research and development for the automotive industry.

The six-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking, if not completely replacing, the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Torino (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

¹"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organized on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided." [1]

- **Cavallo (2020)** [9]: Preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected hospitals.
- **Bertelli (2021)** [10]: Detailed selection of UAV based on mission specifications derived from the medical goods to be transported and ENAC regulatory constraints. Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)** [11]: Identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)** [12]: Development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of an RRT-based algorithm to minimise global risk and flight time.
- **Angelini (2023)** [13] and **Angelini (2024)** [14]: Comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and VTOL aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The present thesis was carried out in collaboration with the three institutions mentioned above to advance the research effort, bridging the gap between their long-term objective and the results achieved so far. This required preliminary work aimed at formalising the project requirements at both a broad, high level and at a specific, first-step level. Having taken a legislative, logistic, market- and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of ENAC (the Italian Civil Aviation Authority - *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained to enable this type of operation. However, as certifications are unavoidably specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst also adopting a "payload-first" approach that diverted the project towards a device for the *safety of the medical payload* rather, than that of the aerial vehicle. The deliberate prioritisation of this aspect is justified in Sections 2.1 and 2.2, where it is illustrated on the basis of a regional market analysis and the exploration of an ongoing paradigm shift in the design process of biomedical devices.

1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Prof. Bianca Giovanardi): Primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): Institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: Provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia S.r.l.**: Industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

1.3. Document Outline

DISCORSIVO. Esempio:

The thesis is structured as follows, first, chapter 2 presents a literature review to outline the several possibilities to predict the buckling load of cylindrical shells and which techniques are available to reduce the sensitivity to imperfections. The chapter concludes by identifying the research gaps related to the buckling of cylindrical shells. From this conclusion, the research objective and scope are defined

in chapter 3. The goal is to include imperfections within the displacement-based Koiter methodology and the necessary steps to achieve this are explained in chapter 4 to 7, as well as how effective this novel modelling technique is. Finally, the concluding remarks can be found in chapter 8 and future research is presented in chapter 9.

2

Background and Research-Development Gap

In technology development, it is important to gain a clear idea of two aspects:

1. **Research aspect:** if the technology that is being developed requires the exploration of an uncharted research niche, as opposed to utilising pre-existing and consolidated knowledge for the creation of a new product;
2. **Development aspect:** whether the proposed technology targets a well-defined market with unmet customer demand and no established supply.

This chapter explores both these aspects in Sections 2.2-2.3 and 2.1, respectively. The aim is to gather sufficient background information, from market allocation of the proposed technology to concrete understanding of airbag design criteria, to depict the choices proposed in Chapter 3 regarding operative *methodology* as an answer to the Research Questions proposed at the end of this chapter.

2.1. Regional Market Analysis: medical UAV logistics in Piedmont

The Italian National Healthcare System (SSN) is structured hierarchically: the guidelines and regulations issued at a *national* level must be implemented locally, at a *regional* level, by each region's *Sistema Sanitario Regionale* (SSR, the Regional Healthcare Service). Vice versa, aviation rules follow European-level enforcement mechanisms: Following the 2019 harmonization of European drone operations across the "open" and "specific" categories [15], operators seeking authorization for high-risk flights are required to conduct a risk assessment using the Specific Operations Risk Assessment (SORA) methodology developed by JARUS¹, making a thorough evaluation of potential failure scenarios while proving compliance with the demanding safety standards imposed by aviation authorities.

include air risk class somehow????? GRC is 6 (p.39). The M2 mitigation, which reduces the effects of ground impact through technical design (p.42, 46), brings it to 3 (p.54). ARC (collisions in airspace) is out of scope.

The discrepancy between *modi operandi* of the aviation and healthcare authorities has shaped the way drone-network operations have mostly been developing in the last decades in Italy: several emerging, albeit scattered, research groups, either cooperating or competing with industrial counterparts, have achieved geographically limited trial deployments while being unable to scale the operation beyond testing.

A complete summary of such operations, limited to the means of investigation available for the present thesis project, can be found in Table A.2 in Appendix A.2. The table shows a clear trend: although potentially booming as several healthcare facilities are exploring the possibility of adopting drone transport and operators are proposing new drone models for them, most activities stop after the initial

¹Joint Authorities for Rulemaking on Unmanned Systems

demonstrations, with the only exception being the region of Abruzzo where the operational phase of the drone drug-delivery service between a strategic hospital and village pharmacies was announced in July 2025 [16], [17]. In general, a pattern of fragmentation, short-lived pilots, and few transitions to a potentially scalable service is observed in the construction of the overall *value chain*, with many independent institutions covering equivalent roles, as shown in Table A.1 in Appendix A.1.

Table 2.1: Actors involved in drone delivery of medical goods in Piedmont, grouped by value-chain stage.

Category	Institution	Scope
Drone manufacturers	SkyEU	Autonomous high-speed eVTOL drones for first aid and critical logistics [18]
Technology partners	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [19]
	FEV Italia	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
Payload / container makers	ABzero (Pisa)	Smart Capsule used for the INDOOR flight: sensorised container for blood, organs and drugs [20], [21]
Operators	ABzero (Pisa)	Supplied the drone and handled the ENAC authorisation for INDOOR [22], [23]
Airspace and infrastructure	Torino Caselle and Cuneo Levaldigi airports	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino	Autonomous flight technologies for transporting biological material (INDOOR) [25]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino)	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost-safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino	INDOOR project partner [19]
End users	AOU Città della Salute e della Scienza	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [21], [24]
	Regional Transplant Centre Piemonte–Valle d’Aosta	INDOOR partner for organ and biological transport [19]
Promoters and funders	Fondazione DOT	Promoter of the INDOOR project [19]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [22]
	City of Turin	INDOOR partner; funded the LastMed call [19], [26]
Regulators and authorities	ENAC	National civil aviation authority; INDOOR partner and flight authorisation [19]
	Regione Piemonte	Regional health authority; drone trials subject to benefit-risk analysis [27]
Alternatives	Tecnocad / Tecnomill / Bylogix	LastMed autonomous ground rover for last-mile drug delivery [28]

Adopting a region-based, rather than a nation-based, approach for market analysis is, therefore, necessary. Table 2.1 highlights stakeholders involved in Piedmont-based operations and their roles in the value chain, illustrating a fragmented scenario: Project INDOOR, launched by Fondazione DOT

in 2021, culminated in a 500 m-long test flight by an autonomous UAV between two Città della Salute hospital sites, carrying medical goods in a dedicated capsule, on 1st October 2024. In this project, several other stakeholders were involved (in fact, most of the ones present in Table 2.1), among which ABzero (a spin-off from *Scuola Normale Superiore di Pisa* responsible for capsule design and handling the ENAC authorisation process for the test flight), parallel research group PIC4SeR from the Department of Electronics and Telecommunications at *Politecnico di Torino*, and MAVTech S.r.l. (UAV systems designer company based in Torino). Independently of the INDOOR project of 2021-2024, the startup SkyEU was founded in 2025 promising to develop eVTOL drone systems for critical logistics within, but not limited to, the healthcare sector.

In parallel, and independently of INDOOR, the research group led by Prof. Enrico Cestino at the Department of Mechanical and Aerospace Engineering (DIMEAS) of *Politecnico di Torino*, together with A.O.U. Città della Salute e della Scienza and FEV Italia, has pursued since 2020 a series of feasibility studies on the Molinette–San Giovanni Bosco route (Section 1.2.2). These covered regulatory and mission evaluation, risk mapping and path planning, payload environmental loads, and cost–safety comparisons between road, drone and eVTOL transport. No flight campaign has been carried out so far. This makes the group the only Piedmontese actor whose work so far has been entirely at the planning and assessment level.

On the other hand, the regional administration’s general stance with respect to this type of operation highlights the fundamental requirement of thorough benefit-risk analyses to guarantee both the safety of the Piedmontese citizens and the preservation of the medical cargo, possibly needing to limit transportation to exceptional emergencies in mountain areas [27]. From an engineering perspective, these criticalities can be translated in the following barriers for the development of a full-scale, UAV-based delivery network:

1. **Authorisation of flights:** Operations beyond the open category are approved by compiling the SORA. Among the different classes of mitigations, *ground risk* mitigations (one of the first steps required in the SORA) reduce the consequences of the UAV’s impact on the ground after a loss of control, for example by introducing a parachute. It is key for technical stakeholders to prove the reliability of their implemented ground risk mitigations to protect civilians in case of drone failure.
2. **Payload integrity and containment:** In accordance with the regional administration’s concerns, current certification standard UN3373 for ground handling of Biological Substances, Category B, establishes that packages must pass a drop test at 1.2m with no leakage from the primary container **CITE STANDARD HERE**. However, in the event of a free-fall from ordinary mission altitudes, this standard loses its significance and the potential for biological material dispersion into the environment becomes highly relevant.
3. **Unsustainability of the current business models** While costs of road and helicopter transport of biomedical goods such as organs, drugs and other critical supplies are expected to fall significantly with drones, most trials are funded by EU projects and one-time grants, with the consequence of making the project unsustainable and non-scalable.

On the basis of this analysis, several research and development gaps can be identified: SkyEU is endeavouring to enact an economically sustainable business model for large-scale UAV networks, but has a small public record; ABzero has patented capsules equipped with electronics that reliably monitor in-flight vibrations and guarantee thermal control of the medical payload, but these are only certified to the UN3373 standard, whereas in case of in-flight failure a much more conservative approach is needed; regional authorities need reassurance on the safety aspect for civilians, as a 500 m test flight on a controlled route shows that the obstacle is not drone-making, but guaranteeing safety in everyday performance rather than in single monitored missions.

In this scenario, the institutions (*Politecnico di Torino (DIMEAS)*, *FEV Italia*, *AOU Città della Salute*) that commissioned the present thesis project aim to tackle, among the three barriers, barrier 2, concerning payload integrity and containment. The proposed solution is an airbag-type device that protects the payload and prevents its dispersion in the event of an impact, especially in view of transporting extremely valuable medicinals and/or biological materials, e.g. human organs for urgent transplant operations. This device is conceived as an add-on to existing capsules such as the one by ABzero, mounted on drones such as SkyEU’s, and is intended to help close the certification gap highlighted

above, while acknowledging that it does not address the issues linked to SORA ground risk or the sustainability of the business model.

2.2. Airframe-first vs. Safety-first: a design-philosophy gap

This section builds on the research-development gap presented earlier by attempting to connect its root causes to a discrepancy between design philosophies applied in the aeronautical and the biomedical sectors. The proposed design solution (the airbag-type device) is, in fact, a suggestion to solve an "aeronautical problem" implementing a "medical approach".

The issue of proving compliance to match demanding safety standards imposed by the certification authorities does not appear to be a Piedmont-only (or even an Italy-only) problem, but rather a widespread trend that affects all non-standard aeronautical projects across the field. As highlighted, for example, in [29], most UAV producers focus on optimising their aerial vehicle's design, minimising mission-related vibrational loads, thermal loads, overall mass, but also proposing ever more efficient cargo-drop mechanisms and ways to overcome geographical obstacles, all partaking in a feedback loop against aerodynamic and flight-mechanics performance metrics. This machine-centric approach appears to be the default, especially considering that UAV-based missions are, naturally, radically different from standard airline and/or general aviation operations, where **the safety of the passengers is undistinguishable from the safety of the vehicle** and where, therefore, stringent certification requirements strongly limit design freedom **CITE SOMETHING HERE?**. A common consequence of focusing exclusively on technological potential when designing UAV-based operations is, therefore, neglecting the safety assurance aspect of the project, which is usually delegated to including a parachute-type device in the final design phases. As pointed out by [30] in their design feasibility evaluation study, the introduction of a parachute usually increases the risk of falling into a mass-increase feedback loop: the parachute increases the airframe's overall mass, which then requires a larger, thus heavier, parachute for efficient deceleration during free-fall, which again results in an excessive mass increase, and so on. This negative feedback loop results either in never reaching design convergence, or otherwise in having to accept a smaller parachute with reduced deceleration capacity, still implying a relatively high ground-impact velocity and almost certainly destructive damage to the airframe, its cargo, and the impacted areas on the ground.

The general impression given by the brief market analysis conducted earlier and the observations gathered in the present paragraph is that, perhaps, the "airframe-first" approach adopted in the majority of **nascenti? appena nuove?** UAV-based operations is not sustainable for the projects' longevity beyond pilot flights, and that it might be necessary to embrace a paradigm shift in design approaches. Specifically, there might be potential in drawing inspiration from the biomedical field, where medical technology designers are encouraged to shift from a techno-centric to a human-centred approach to avoid the *translation gap* or *implementation gap*, i.e. the potential to introduce or enhance health inequity when trying to implement extremely advanced technology that does not take into account all necessities and limitations of human patients [31].

Just like success is measured on the patient and not on the device in the biomedical field, in the case of UAV-based medical delivery operations success should be measured on payload integrity and protection of civilians, not (only) on the capabilities of the airframe. A drone that survives an in-flight failure while sacrificing its payload and threatening irreversible human damage has undoubtedly failed its mission, as for a biomedical device that functions correctly but does not serve the patient. This reasoning leads to the first of the contributions proposed in the present thesis, which is the "*safety-first*" design principle presented in Section 3.2.1. This thesis aims to serve as a test-bed for this proposed approach, where exploring the airbag design process helps bridge the gap between current development efforts and certification.

2.3. Airbag impact attenuation: background and design variables

Airbag-type devices have a well-established, enduring research-development record with contributions found across multiple industries. The information presented in Sections 2.3.1 and 2.3.2 aims to represent the State of the Art practices adopted in airbag design. The scope of this investigation is developing a solid, although perhaps not comprehensive, understanding of design recommendations on which to

base the proposal for the design workflow of later Section 3.2.2.

2.3.1. Established applications and common working principle

Airbag technology is well established across two primary sectors: *automotive* and *aerospace*.

In the automotive industry, these systems are implemented during vehicle crashes, distinguishing between general automobile crashes and motorcycle impacts. In the space industry, ever since the early space missions of the 1960s, airbags have been deployed during the final landing phases of space capsules. Within the aeronautical industry, specifically in the "specific" operations category, other niche applications have emerged, such as the experimental study of adaptive airbag systems for the protection of UAVs and general aviation vehicles [32] (following a long track record of exploration in the field of adaptive technologies for impact absorption [33], [34]).

In either case, the primary objective of the airbag-type device remains the same: to be stored compactly in ordinary mission conditions, inflate within the allowable time before impact (whether during an emergency impact or landing), and decelerate the target body (whether a crash test dummy or a spacecraft) far more gently than in case of direct impact. When analysing these systems, purely analytical, hard-codable solutions are typically limited to single-degree-of-freedom (1-DoF) formulations, immediately resolving to the use of far more computationally-heavy finite element analyses to capture the complex two- and three-dimensional effects of the impact. The only exception to this trend is the technical report NATICK/TR-87/042 published by T. F. Patterson in 1985², which presents a 6-DoF analytical simulation built on closed-form component models, but remains inaccessible at the time of writing of the present literature search.

Despite the diversity of fields, all applications share identical fundamental working principles. An example of an analytical study of airbag-based impact events, and one of the most significant papers in shaping this thesis' work, is Zhou et al. (2019) [35], in turn heavily rooted in the SAND93-2133 technical report [36] on a code for predicting the performance of the Pathfinder lander's airbag impact attenuation system, which describes the core working principle of airbag energy dissipation in a one-degree-of-freedom impact event.

MORE DETAILED DESCRIPTION HERE NEEDED? YES remake side-by-side airbag impact diagram? yes, to explain the working principle of the CALCULATION LOOP

The idea is that i can turn this working principle into a multi-variable problem that requires some optimisation too.

2.3.2. Design guidelines and variables

Variabili da prendere in considerazione quando c'è da progettare un airbag

Size Venting mechanism Fabric/material (bursting vs. venting aspects) Impact modelling (horizontal velocity component, mention that there is a study that was not accessible to analytically include it, everyone just resorts to FEM)

Air injection - pirotecnico vs. non, differenti meccanismi (polverina innescata, oppure bombolette simili a quelle motociclistiche)

Patterning - transforming the 3d-shape into 2d seamable shapes, minimising number of seams. Seaming - where to put them and how to stitch them, with different patterns possible. Usually simulatable with beam elements Folding patterns and techniques - some standard patterns that need to be adjusted according to local geometry and OPTIMISED for deployment time. Housing/container - unfolding triggering the opening of the container.

Design ideas / process 8 Optimisation and evaluation 14 Testing validation 15 1DoF case – things to remember 16

²T. F. Patterson, *Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers*, NATICK/TR-87/042, U.S. Army Natick Research, Development and Engineering Center, Aug. 1985.

Summary of design variables

Based on the information gathered in Section 2.3.2, it is possible to summarise airbag design variables and classify them based on the design phase they partake in, as shown in Table 2.2. Specifically, Phase 1 focuses on designing an airbag in its fully-inflated configuration to limit peak impact acceleration below the allowable limit set by the payload's integrity requirement, while Phase 2 requires making design choices oriented at optimising deployment time and leading to prototyping.

Airbag design variables	
Phase 1 - design for peak acceleration	Phase 2 - design for deployment time
1. Shape 2. Characteristic size 3. Inflation pressure 4. Fabric type, thickness 5. Venting mechanism - vent burst pressure and/or fabric porosity	1. Housing/container size and geometry 2. Air injection inlet pressure 3. Air injection mass flow rate 4. Seams locations 5. Folding pattern 6. Number of folds

Table 2.2: Airbag design variables by design phase.

2.4. Research-Development Objective

Having presented the context in which the research project is allocated and having gathered the necessary design information, it is possible to address the requirements set by the project commissioners:

- Identify and characterise the mechanical allowables of typical medical payloads based on general packaging requirements for UAV-transported medical goods.
- Develop an informed conceptual design of an airbag system capable of deploying during emergency release conditions to absorb impact energy and preserve payload integrity.

These requirements can be summarised in the following Research Objective and Research Questions:

Research-Development Objective
Develop and validate a design workflow to model and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the primary objective to ensure the structural integrity of pharmaceutical materials and their containment during drone failure and emergency payload release, thereby preventing environmental dispersion and supporting certification requirements.
Research Question(s)
How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events? 1. What are the allowable mechanical limits (acceleration, stress, etc.) for medical products during emergency release, and how should these define the design envelope? 2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements? 3. How can the optimised inflated geometry be transformed into a compact and functionally deployable system?

Methodology - Airbag Design Workflow

3.1. Problem statement

To properly outline and, more importantly, test the workflow proposed as a main objective of this thesis, it is necessary to identify the specific context in which this project is embedded, as it originates from prior research work conducted within the specific framework of Torino. The design inputs presented below were gathered from analysing the results presented by the group ([9], [10], [11], [12], and [13]) and categorising them into mission aspects on which to root the proposal of Section 3.2.

3.1.1. Reference mission and design inputs

Route and flight profile

The reference mission is based on a hospital-to-hospital route as initially suggested by Cavallo (2020) [9], connecting two hospitals across the city of Torino as shown in Fig. ?? **INCLUDE FIGURE HERE!!!!**, representing a city map with route schematics. In particular, the starting point is the Molinette Hospital in the south-east region of the city, and the destination is the San Giovanni Bosco hospital in the **north-west** sector. As they are located far apart, these two hospitals require medical deliveries to traverse a broad part of the city's territory. The hospitals are connected through two possible routes:

- **Route A:** Initially hypothesised by Cavallo and confirmed by Bertelli (2021) [10], it takes advantage of the above-water air corridor over the river Po and its tributary Stura, mostly skirting the city and flying over city ground only in the relatively short segments connecting the hospitals to the rivers. Among the two routes, it is the safest (and, thus, most certifiable) as it reduces the probability of ground impact in case of drone failure due to the UAV's permanence in the above-water air corridor. This route was selected as viable within the Concept of Operations (ConOps) that informed the preliminary drafting of the SORA. Approximate length: 10 km. Approximate duration time: 15 min, comparable to ambulance-based delivery times.
- **Route B:** A crude alternative to Bertelli's Route A, connecting the two hospitals in a straight line over the city. Angelini (2023) [13] deemed this route infeasible based on their comparative safety analysis of delivery means, primarily due to the lack of proper risk mitigation measures concerning in-flight failures. However, as the approach proposed in this thesis aims at partially mitigating these risks, it was chosen to consider this route (at least for preliminary evaluation).

In either case, as specified in Bertelli's ConOps, the cruising height is about 75 m, at 17 m/s cruising velocity (approximately 70% of the maximum cruising speed allowed by the UAV platform of 25 m/s, unsustainable for the entire duration of the mission at the time of evaluation), with possible speed variations according to mission urgency if specified in the ConOps as considered by Angelini.

Accounting for the presence of wind in the operating environment, wind speed in Turin averages about 2 m/s (7 km/h), with a peak of about 28.4 m/s (102 km/h) in 2013<empty citation> **THE REPORT**

ABOUT WIND FROM THE COMUNE!!!!. go and look AT THE CODE FOR THE VALUE OF WIND YOUR DUMB ASS DECIDED TO USE

EXTRA THINGS TO INCLUDE ENAC's restrictive rules are satisfied by the water corridor (Cav p.37) Risk-based route selection (Ang, Fio): Dijkstra and A* for the minimum-risk path (Ang p.17). It is a static algorithm run on a per-cell risk map, and the output is maps for the Turin province. Input maps: population density (p.35) and shelters (ref. 17, p.40) Probabilistic risk maps Fio 2021 uses RRT* for path planning, with the "layers" idea of combined risk factors Primatesta (Bert refs 8–9, Fio p.40) was an inspiration for Davide's work too. Romeo and Cestino is Ang ref. 8. Merit index: three conditions (payload, route length, safety), with weights that depend on mission type (Ang p.86). Safety penalises the drone much more heavily than the eVTOL.

UAV Platform

The UAV platform choice is inherently linked to the reference mission specifications above, meaning that the platform is required to sustain the cruising speed requirements specified in the ConOps. A Maximum Takeoff Mass (MTOM) of 9 kg and a 0.81 m characteristic dimension were also specified in Bertelli's ConOps, leading to the suggestion of a potential drone model suitable for the situation: DJI's *Matrice 300 RTK*. Despite having characteristics suitable for the speed requirements, the drone has a payload limit of 2.7 kg and a wind resistance of 15 m/s in cruise. While this drone was initially selected for the preliminary SORA, its specific parameters are not binding for the present study. In particular, the payload-mass and wind-resistance limitations may be disregarded, given that the ConOps set a much broader requirement for MTOM. In addition to these requirements, Cavallo, in their UAV platform market evaluation, considered several other factors: maximum flight time (usually corresponding to half the autonomy), geometry, glide speed, and frontal area, the latter covering an important role in free-fall drag calculations. By ConOps, the platform is equipped with a parachute, whose contribution is ignored in the calculations regarding the free-fall event.

Failure scenario and impact conditions

The chosen failure scenario implies an in-flight malfunction or fatal failure of the UAV in combination with a failure of the parachute mounted on the drone by ConOps. Excluding the nominal, non-conservative parachute descent of 3 m/s estimated by Bertelli, in this extreme failure scenario the payload is dropped in free air, separated from the drone at an altitude of 75 m above ground. After the failure, a *ballistic* descent mode is chosen for calculations of impact time and point depending on the release position on the map, including the effects of gravity and aerodynamic drag (as opposed to *uncontrolled glide* descent mode, which is applicable to the UAV platform but not to its detached payload), used in Angelini's impact probability study for ground risk evaluation and inspired by PRIMATESTA CHECK!!! Wind enters the calculations (as mentioned in the reference route definition) as a horizontal velocity component.

Payload container

The design of the payload container to be carried by the UAV platform in delivery missions was proposed, but never fully prototyped, by Sivieri (2021) [11]. Its geometry is shown in Fig.?? INSERT FIGURE HERE: The central body is a cylinder capped with two hemispherical domes. In Sivieri's original design, this container had to be constructed via stratifying four layers, from inside out: first layer of 3d-printed PET-G with thickness 2 mm, second layer of manually applied thermal insulation of unknown effective thickness, third layer of 3D-printed PET-G and 2 mm thickness, and a final layer of manually-applied thermo-reflective paint. This sandwich-type structure was designed specifically for evaluating the thermal and vibration loads of a reference mission similar to the one presented earlier, and thus avoiding making considerations on transportability, attachment mechanisms, and other application-specific parameters.

Due to the artisanal nature of this design, linked to the difficulty in modelling specific 3D-printed PETG and the unquantifiable specifications of the insulation material, the payload container was modelled by retaining its shape with a 5 mm thick shell [verify thickness in the FE model] of injection-moulded Acrylonitrile butadiene styrene (ABS), commonly used in packaging. The cavity is filled with polyurethane foam, chosen to simulate a protective wrapping (such as bubble wrap) that medical operators would use as established praxis to mitigate vibrations. The thermal paint is omitted from the payload container modelling process.

Payload

Considering the wide variety of medical goods potentially suitable for UAV-based air transport, it was necessary to identify a payload representative of standard shipments, excluding human organs and specialized equipment (e.g. defibrillators) to focus specifically on pharmaceuticals. Therefore, glass phials of the medicinal product *Akineton Fiale*, containing *biperidene lactate* at a concentration of 5 mg/ml in injectable solution and shown in Fig.3.1¹, were chosen. The product is commercially supplied in boxes of five phials, each individually packaged in a plastic blister tray.

Akineton is a pharmaceutical product commonly used in the treatment of Parkinson's disease and drug-induced extrapyramidal symptoms, and is usually administered by intravenous or intramuscular injection. For the purposes of the present thesis, only the mechanical properties of the glass container are of interest, as the pharmacological action of the medical fluid is not relevant to the type of structural analyses implemented in this study.

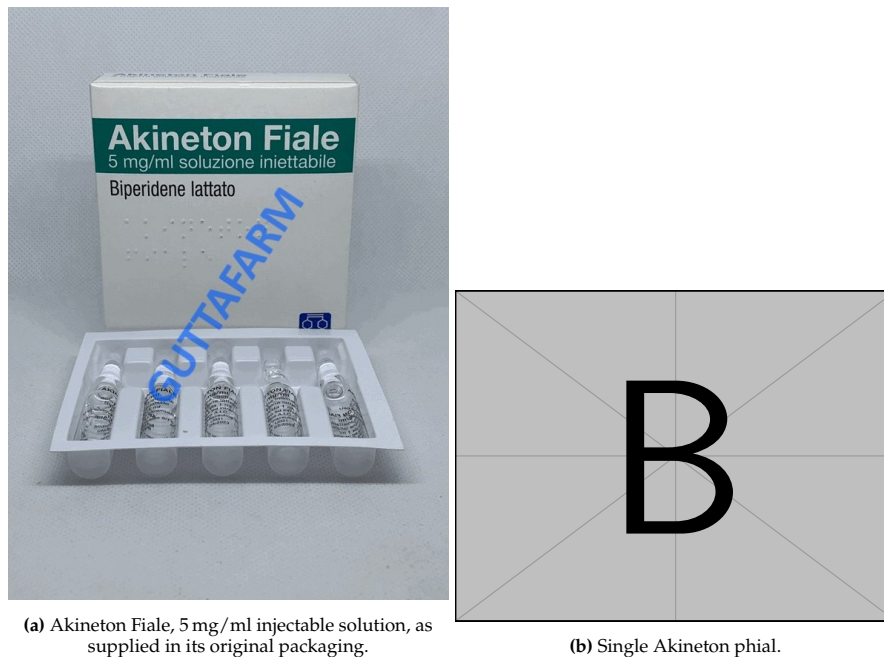


Figure 3.1: Akineton Fiale packaging and single phial.

3.1.2. Information gap

The only missing inputs required for a full design workflow evaluation are:

- The derived *impact velocity*, angle, and mass for the selected scenario;
- The quantification of the phials' fragility through the characterisation of their material's mechanical *allowables*;

These failure scenario data must be fed into the design loop(s) proposed below to enable an effective design of the airbag-based impact attenuation system.

3.2. Proposed approach

3.2.1. Outer design loop: formalising the Safety-First principle

In response to the gap identified in Section 2.2, the first contribution of this thesis is to propose an alternative design approach adopting (and later testing) the following guidelines:

¹Source of photo: <https://www.guttafarm.net/akineton-cena>.

Safety-First Design Principle

1. The airframe and payload must be considered as two distinct entities;
2. In the type of operations considered, saving the payload takes higher priority than saving the airframe (carrying organs, expensive vaccines, emergency medicines, or high-value medical equipment is worth more than the drone itself);
3. Overall, the objectives are to minimise or mitigate:
 - ground impact consequences,
 - payload damage (preserving its integrity),
 - environmental dispersion;
4. These considerations should be applied at the beginning of the design process, possibly by integrating multiple safety devices (e.g., an airbag and a parachute working in tandem) and adjusting flight-mechanics considerations accordingly.

A direct consequence of applying the Safety-First design principle in UAV-based operations' design is changing the sequence of design events, as exemplified in the simple flowcharts of Fig. 3.2, where Fig. 3.2a and Fig. 3.2b compare scenarios in which the mission loop can be interchanged with the safety loop to achieve design convergence.

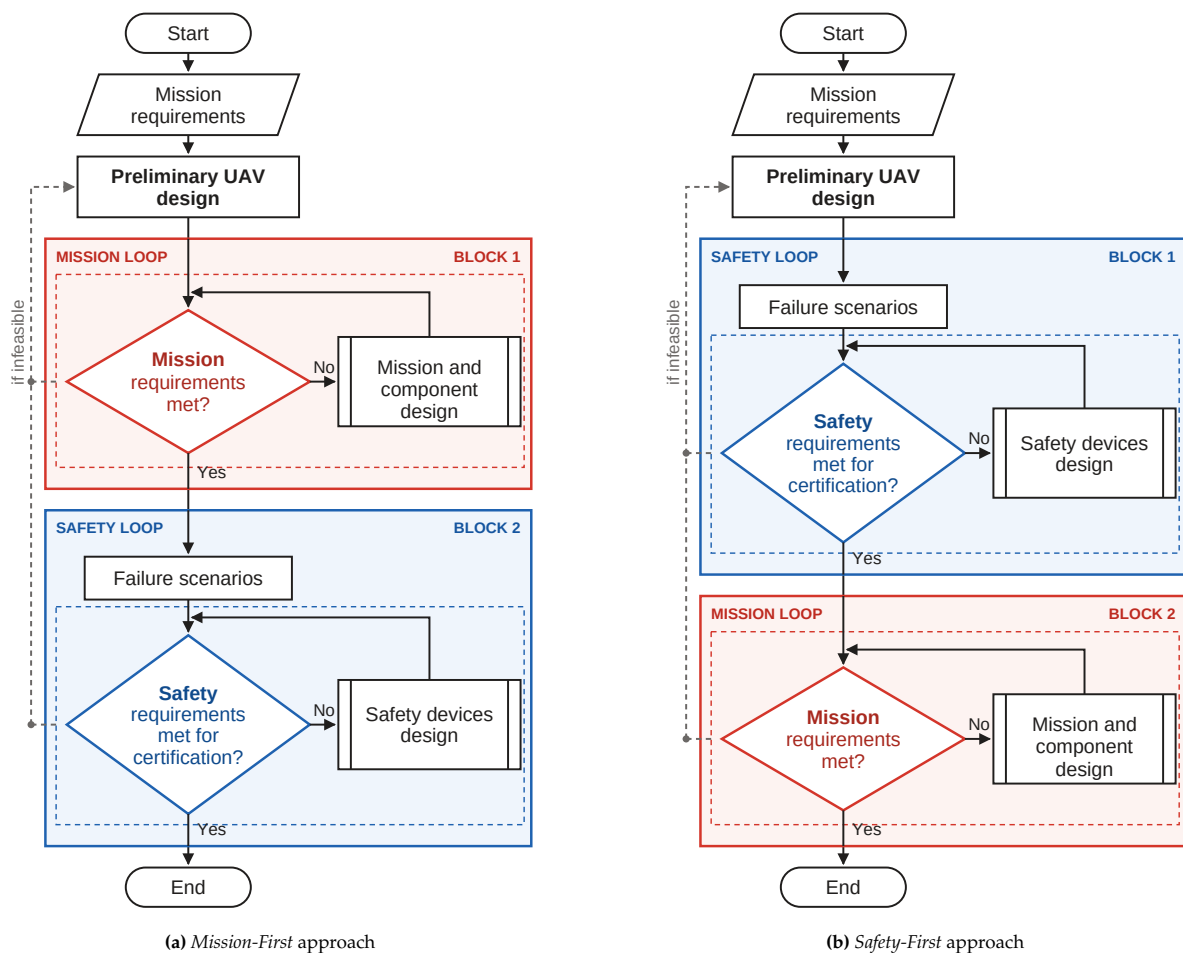


Figure 3.2: Sequence of design events in the two approaches: the order of the mission and safety loops is swapped.

3.2.2. Inner design loop: a payload-protection subsystem design workflow

* Inner design loop formalisation: specific requirements and design steps of one of the processes in the safety loop of 3.1b. DEVELOPING, greater deal of a process. * the loop is payload-specific and device-specific, i.e. an airbag needs to protect a glass phial. * The diagram + objectives, i.e. where this MSc Thesis “stops”

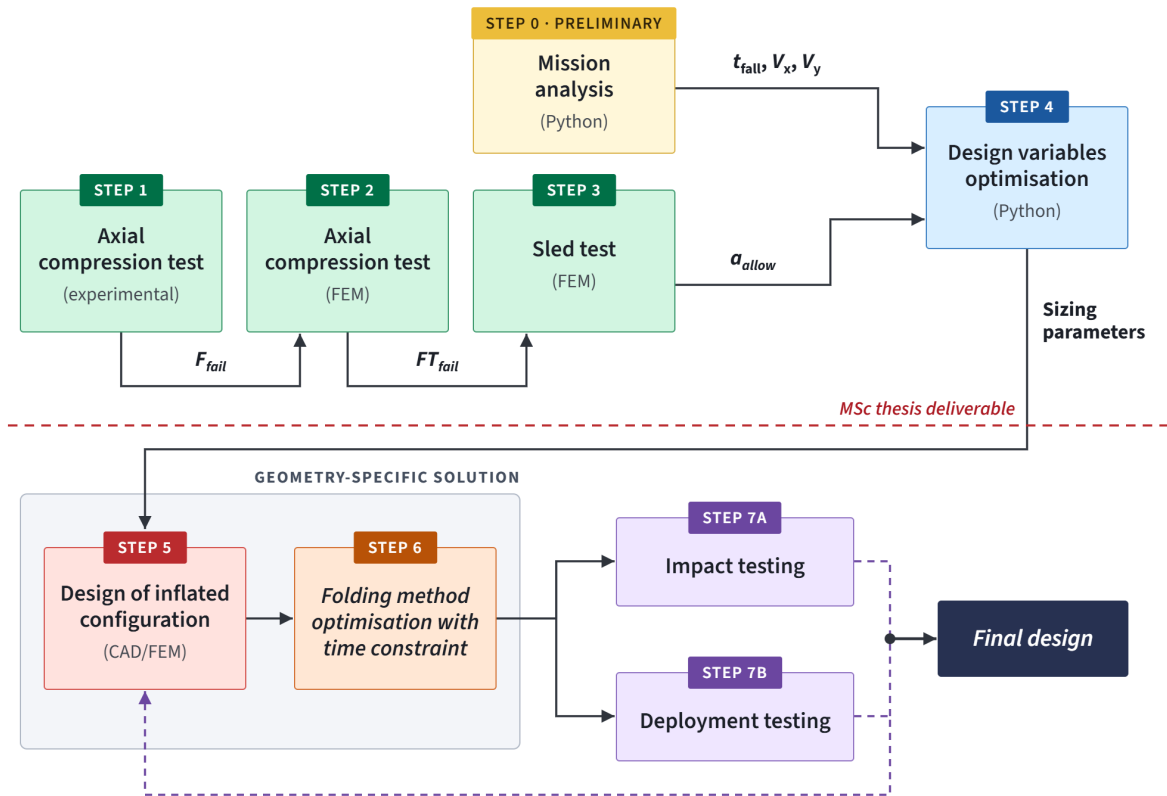


Figure 3.3: Inner design loop, specific for the airbag-based payload-protection subsystem.

The upper row (Steps 0–4) gathers the mission inputs (t_{fall} , V_x , V_y) and the payload allowable acceleration a_{allow} , derived through experimental and numerical axial compression tests (F_{fail} , FT_{fail}) and an FEM sled test, and feeds them into the optimisation of the design variables. The resulting sizing parameters pass to the geometry-specific solution in the lower row (Steps 5–6), which is verified by impact and deployment testing (Steps 7A–7B) and iterated (dashed line) until the final design is reached. The red dashed line marks the scope of this MSc thesis.

0. **Mission analysis:** calculate impact xxxsxsxsxsxs
1. **Axial compression test (experimental):** experimental characterisation of glass phials breaking point
2. **Axial compression test (FEM):** matching the experimental behaviour at a fem level to reproduce mechanical behaviour of material, limited to the compression behaviour as that is what is going to happen in the event of a fall.
3. **Sled test** - emulation of a sled test inspired by the automotive field, input-output
4. **Design variables optimisation** - code, trying to simulate the impact of the actual thing in python to seize the airbag
5. **Design of inflated configuration** - tailoring the airbag seizing to the payload and UAV geometry
6. **Folding method optimisation with time constraint** - matching gas generator mechanism to geometry of the drone and payload and impact time and deployment requirements (really hear?), and including folds in the idea, like, standard folds, etc.

7. **Impact and Deployment testing** validating the prototype through testing in a feedback loop where virtual and practical behaviours match.

* bring it back to the design variables found in section 2, explicitly saying which ones i'll focus on and why - payload-specific, ect.

CHOOSE LOOKING AT THE CODE!!!!!!!!!!!!!!!!!!!!!!

for the last part of the diagram Phase 1 variables for - acceleration 1. Shape 2. Size 6. Inflation pressure 3. Fabric 4. Vent size - venting mechanism (hypothesis) / Vent burst pressure

SOME ARE MISSING HERE - check the code

Payload-specific variables - deployment time 1. Housing/container size and geometry 2. Air injection inlet pressure 3. Air injection mass flow rate 4. Seams locations 5. Folding pattern 6. Number of folds

THESE SHOULD BE LISTED ONLY FOR PROTOTYPING.

4

Results - Through the Workflow

Do not include too many implementation details

4.1. Mission and failure scenario analysis: calculating V_x and V_y

* Mi servono degli input: prendo le informazioni raccolte nella fase preliminare di analisi missione (ref analisi missione) per stabilire probabilità di impatto and what not.

* traiettoria * punto di partenza, punto di arrivo * drone scelto * payload scelto * medicinali scelti per simulazione

* serve energia di impatto?

—> risultati del WP2

4.2. Studying the payload: calculating allowable impact acceleration

4.2.1. Material characterisation

TEST

FEM

MENTION MODEL ITERATIONS TRIAL-AND-ERROR IN THE APPENDIX B!!!! FROM ABAQUS IMPLICIT TO CURRENT DECISIONS. Step by step, as in "version 1, abaqus implicit". Version 2, ls-dyna tetra elastic. Etc. MODAL PERTURBATION

* impact of container in LS-Dyna and what not. * diversi angoli di assetto per vedere il caso peggiore di accelerazioni locali * cosa possono sopportare le fiale e il contenitore???? * tipi di impatto, contenitore tipico che le contiene (menzionare la tesi del tipello) e che per il dimensionamento non viene considerato, approssimato a infinitamente rigido e che trasmette le accelerazioni di impatto così come le riceve —> airbag deve essere dimensionato così —> tutte le analisi statistiche di prima, che hanno tenuto in considerazione anche il fatto che prima o poi si sorvolerà la città una volta risolti i cavilli burocratici che impongono una certa mitigazione del rischio —> airbag.

—> risultati del WP3

—> CONCLUSIONE: I have what I need to design the damn airbag.

4.2.2. Sled test

4.3. Design variables optimisation: the code!!!

* codice ispirato da Zhou et al 2019 - spiegazione del modello e dell'aspetto dei vari blocchi * Sapendo che le thresholds delle provette sono tot, allora si può convergere verso un pre-dimensionamento dell'airbag

con tot dimensioni o caratteristiche. * oracle code for direzionamento verso un punto di partenza per il design

* I CHOSE THE CRUDE IMPLEMENTATION — THIS IS BECAUSE THE ONLY COUPLING ANALYTICAL MODEL IS MISSING AND I CANNOT RETRIEVE IT. THEREFORE THIS IS MY IMPLEMENTATION XXX Title: "Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers"

Author: Timothy F. Patterson

Report: NATICK/TR-87/042 (DTIC accession ADA199593)

Date: August 1985 (work performed March 1984–May 1985)

4.4. Summary of results

Actual design variables p inflation, diameter, shape, etc

****RESULTS: BRING IT BACK TO THE RESEARCH QUESTIONS!!!!****

5

Conclusions

5.1. General summary

Bring back the idea of the safety-first workflow into the broader context of the long-term, drone-network-all-over-Italy vision.

5.2. Recommendations for future work

Feeding these results in the “outer” design loop, updating the hardware and mission design, re-evaluating initial conditions to re-inform the “inner” design loop -> once reached convergence, the need to simulate and test the design implemented on the drone, with informed decisions on inflation mechanism and folding method.

combination of parachute (for, let’s say, half the falling height) and then airbag for the second half? the objective is to minimise overall mass.

we are not done at all. There is the geometry-specific part that is required, including the design recommendations in Section 2.3.2 and going on, PAST the workflow, as mentioned in Section 3.2.2

6

Project Management

Inspired by the Gibb's reflective cycle (?)

6.1. Strategy and planning

planning strategy (especially at the beginning), 5 supervisors to update independently, different means of updates - weekly meetings that became only when i had updates - regular bi-weekly meetings - irregular meetings and informal discussions here and there because of huge workload

6.2. Challenges

6.2.1. challenge 1: aligning interests for thesis scope

- 5 supervisors with 5 different visions of my thesis work. what contributed was a lack of a unified platform to check my progress and congregate materials.

6.2.2. challenge 2: Computational resources NO MORE

- by contract, politecnico should have given the resources, hpc and licences - unexpected maintenance, with delay - Bianca Ex Machina, access to DelftBlue. Out of contract, otherwise the thesis should have been changed

6.2.3. challenge 3: IT infrastructure bottleneck

- resulting project flow from company pc to personal pc, because no permission to install a VPN and launch from the company's computer.

6.2.4. challenge 4: going against the judgement of supervisors

- scegliere shell invece che solids in case of fev supervisor - scegliere full invece che reduced model in case of tudelft supervisor.

6.3. Lessons learned

I'm really mad

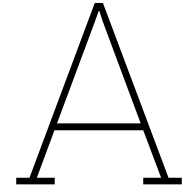
References

- [1] G. Fantato et al., "Italian territorial emergency medical services - 118: A type of advanced EMS vehicle in the experience of the Piedmont region," *Ital. J. Emerg. Med.*, vol. 12, no. 2, pp. 81–89, Aug. 2023. doi: 10.23736/S2532-1285.23.00185-4
- [2] N. Abrescia, A. Maddaloni, G. Molinaro, A. Capponi and M. D'Abbraccio, "The long crisis of Italy's National Health Service: Universalism on paper, inequity in practice," *Eur. J. Public Health*, vol. 36, no. 4, ckag138, Aug. 2026. doi: 10.1093/eurpub/ckag138
- [3] L. Garattini, M. Zanetti and N. Freemantle, "The Italian NHS: What lessons to draw from COVID-19?" *Appl. Health Econ. Health Policy*, vol. 18, no. 4, pp. 463–466, Aug. 2020. doi: 10.1007/s40258-020-00594-5
- [4] M. Capuzzo, G. L. Viganò, C. Boniotti, L. M. Ignoti, C. Duri and V. Cimolin, "Impact of the first phase of the COVID-19 pandemic on the acquisition of goods and services in the Italian health system," *Int. J. Environ. Res. Public Health*, vol. 19, no. 4, p. 2000, Feb. 2022. doi: 10.3390/ijerph19042000
- [5] M. Balasingam, "Drones in medicine—The rise of the machines," *Int. J. Clin. Pract.*, vol. 71, no. 9, e12989, Sep. 2017. doi: 10.1111/ijcp.12989
- [6] J. E. Scott and C. H. Scott, "Drone delivery models for healthcare," in *Proc. 50th Hawaii Int. Conf. Syst. Sci. (HICSS)*, Jan. 2017, pp. 3297–3304. doi: 10.24251/hicss.2017.399
- [7] E. B. Odigie, B. I. G. Adejumo and A. E. Oigbochie, "Importance of drones in healthcare delivery amid a pandemic: Current and generation next application," *Open J. Med. Res.*, vol. 2, no. 1, pp. 1–13, 2021. doi: 10.52417/ojmr.v2i1.187
- [8] A. Gangwal, A. Jain and S. Mohanta, "Blood delivery by drones: A case study on Zipline," *Int. J. Innov. Res. Sci. Eng. Technol.*, vol. 8, no. 8, pp. 8760–8766, Aug. 2019. doi: 10.15680/IJIRSET.2019.0808063
- [9] F. Cavallo, "Innovative ways of distribution for life-saving medicines during emergency situations," Master's Thesis, Politecnico di Torino, Torino, Italy, 2020.
- [10] D. Bertelli, "Studio di fattibilità di un sistema innovativo, tramite drone, per il trasporto di farmaci salvavita in situazioni di emergenza," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [11] F. Sivieri, "Valutazione degli effetti ambientali sul trasporto di materiale biologico e farmaci tramite aeromobili a pilotaggio remoto," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [12] F. Fiorentino, "Path planning strategies for drone delivery for life-saving pharmaceuticals," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [13] D. Angelini, "Progettazione e valutazione di un sistema di trasporto innovativo aereo per farmaci emoderivati in condizione di emergenza," Master's Thesis, Politecnico di Torino, Torino, Italy, 2023.
- [14] D. Angelini, E. Cestino, D. Cestino and F. Cattel, "Comparative analysis of eVTOL, drone and ground transportation systems for emergency delivery of blood-derived medication," in *Proceedings of the International Council of the Aeronautical Sciences (ICAS)*, Florence, Italy: ICAS, 2024.
- [15] European Commission, *Commission implementing regulation (EU) 2019/947 of 24 May 2019 on the rules and procedures for the operation of unmanned aircraft*, Official Journal of the European Union, vol. L 152, pp. 45–71, Jun. 2019. [Online]. Available: http://data.europa.eu/eli/reg_impl/2019/947/oj
- [16] *Farmaci consegnati con i droni: Al via la fase operativa del progetto Telespazio-Asl*, ASL 1 Avezzano Sulmona L'Aquila, Accessed 5 October 2026. [Online]. Available: https://www.asl1abruzzo.it/archivio2_notizie-ed-eventi_0_1191.html
- [17] *Farmaci via drone ai pazienti fragili: Primo volo da Avezzano a Collelongo*, Abruzzo Speciale, Accessed 5 October 2026. [Online]. Available: <https://www.abruzzoSpeciale.it/2025/08/01/farmaci-con-drone-avezzano/>

- [18] *SkyEu*, SkyEU website, Accessed 5 October 2026. [Online]. Available: <https://skyeu.it/>
- [19] *A Torino vola il drone per il trasporto di organi e materiale biologico*, Fondazione D.O.T. Onlus, Accessed 5 October 2026. [Online]. Available: <https://www.fondazionedot.it/a-torino-vola-il-drone-per-il-trasporto-di-organi-e-materiale-biologico/>
- [20] *Il sistema ABzero*, ABzero website, Accessed 5 October 2026. [Online]. Available: <https://www.abzero.it/il-sistema-abzero/?lang=en>
- [21] *A Torino vola il drone per il trasporto di organi e materiale biologico: Al via la sperimentazione, con tragitto dal CTO alle Molinette*, AOU Città della Salute e della Scienza di Torino, press release, Accessed 5 October 2026. [Online]. Available: https://www.cittadellasalute.to.it/index.php?option=com_content&view=article&id=24515
- [22] *Il progetto INDOOR: Nuove prospettive per il trasporto di organi grazie alla tecnologia dei droni*, Fondazione D.O.T. Onlus, Accessed 5 October 2026. [Online]. Available: <https://www.fondazionedot.it/il-progetto-indoor-nuove-prospettive-per-il-trasporto-di-organi-grazie-alla-tecnologia-dei-droni/>
- [23] *A Torino vola il drone per il trasporto di organi e materiale biologico*, TecnoMedicina, Accessed 5 October 2026. [Online]. Available: <https://www.tecnomedicina.it/a-torino-vola-il-drone-per-il-trasporto-di-organi-e-materiale-biologico/>
- [24] *Anche a Torino un progetto per decarbonizzare la logistica sanitaria con droni a idrogeno*, HydroNews, Accessed 5 October 2026. [Online]. Available: <https://www.hydronews.it/anche-a-torino-un-progetto-per-decarbonizzare-la-logistica-sanitaria-con-droni-a-idrogeno/>
- [25] *A Torino vola il primo drone per il trasporto di organi e materiale biologico*, Politecnico di Torino, Poliflash, Accessed 5 October 2026. [Online]. Available: <https://www.polito.it/ateneo/comunicazione-e-ufficio-stampa/poliflash/a-torino-vola-il-primo-drone-per-il-trasporto-di-organi-e>
- [26] *LastMed: Il robot che consegna farmaci agli anziani*, New Pharma Italy, Accessed 5 October 2026. [Online]. Available: <https://www.newpharmaitaly.it/lastmed-il-robot-che-consegna-farmaci-agli-anziani/>
- [27] *Idroni e il delivery sanitario*, Consiglio Regionale del Piemonte, press release, Accessed 5 October 2026. [Online]. Available: <https://www.cr.piemonte.it/cms/articoli/comunicati-stampa/i-droni-e-il-delivery-sanitario>
- [28] *Tecnocad, farmaci a persone fragili da rover robot automatizzato*, ANSA, Accessed 5 October 2026. [Online]. Available: https://www.ansa.it/piemonte/notizie/2026/03/23/tecnocad-farmaci-a-persone-fragili-da-rover-robot-automatizzato_b266e9c6-f168-4b25-a2e7-775a83795571.html
- [29] R. K. Chhetri, S. Jetawatthana and T. Khamvilai, "A survey of medical drones from flight dynamics, guidance, navigation, and control perspectives," *arXiv preprint arXiv:2602.06969*, 2026. arXiv: 2602.06969 [cs.R0].
- [30] V. Alulema, G. Acosta-Vargas, C. Guevara and P. Acosta-Vargas, "How stakeholder requirements constrain drone-based medical delivery: Probabilistic sizing of evtol aircraft at high altitude," *Scientific Reports*, vol. 16, no. 1, p. 26444, 2026. doi: 10.1038/s41598-026-54743-2
- [31] N. M. Rodriguez, G. Burleson, J. C. Linnes and K. H. Sienko, "Thinking beyond the device: An overview of human- and equity-centered approaches for health technology design," *Annual Review of Biomedical Engineering*, vol. 25, no. 1, pp. 257–280, 2023. doi: 10.1146/annurev-bioeng-081922-024834
- [32] J. Holnicki-Szulc et al., "Adaptive airbag systems for protection of general aviation," in *Proc. 2nd ECCOMAS Thematic Conf. Multidisciplinary Des. Optim. Aerosp. Syst. (AeroBest 2023)*, Lisbon, Portugal, Jul. 2023.
- [33] J. Holnicki-Szulc, C. Graczykowski, G. Mikułowski, A. Mróz, P. Pawłowski and R. Wiszowaty, "Adaptive impact absorption—the concept and potential applications," *Int. J. Protective Struct.*, vol. 6, no. 2, pp. 357–377, Jun. 2015. doi: 10.1260/2041-4196.6.2.357

- [34] J. Holnicki-Szulc, C. Graczykowski, G. Mikułowski, A. Mróz and P. Pawłowski, "Smart technologies for adaptive impact absorption," *Solid State Phenom.*, vol. 154, pp. 187–194, 2009. doi: 10.4028/www.scientific.net/SSP.154.187
- [35] X. Zhou, S. M. Zhou and D. K. Li, "Optimal design of airbag landing system without rebound," *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 531, no. 1, p. 012001, 2019. doi: 10.1088/1757-899X/531/1/012001
- [36] J. K. Cole and D. E. Wayne, "BAG: A code for predicting the performance of a gas bag impact attenuation system for the PATHFINDER lander," Sandia National Laboratories, Albuquerque, NM, USA, Tech. Rep. SAND93-2133, Nov. 1993.
- [37] *Trasporto di sangue con droni cargo sanitari: Il progetto Aetos arriva a Ischia*, DiSalute, Accessed 5 October 2026. [Online]. Available: <https://disalute.it/trasporto-sangue-droni-cargo-sanitari-aetos-ischia/>
- [38] *Droni a idrogeno per il trasporto di medicinali dall'Aeroporto di Padova: Ecco il progetto Sandbox!* Hydrogen-news.it, Accessed 5 October 2026. [Online]. Available: <https://hydrogen-news.it/droni-a-idrogeno-dallaeroporto-di-padova-ecco-il-progetto-sandbox/>
- [39] *Droni, boom del trasporto in volo di farmaci*, Adnkronos, Accessed 5 October 2026. [Online]. Available: https://www.adnkronos.com/droni-boom-del-transporto-in-volo-di-farmaci_4BrYXkMkpLBoedJwwIB6nw
- [40] *Defibrillatore? Lo porta il drone*, Tecnica Ospedaliera, Accessed 5 October 2026. [Online]. Available: <https://www.tecnicaospedaliera.it/defibrillatore-lo-porta-drone/>
- [41] *Droni-infermiere: Il boom del trasporto in volo di farmaci, sangue e organi*, Meteoweb, Accessed 5 October 2026. [Online]. Available: <https://www.meteoweb.eu/2022/11/droni-infermiere-boom-transporto-farmaci-sangue-organi/1001164442/>
- [42] *Il defibrillatore del 118 arriva con il drone*, ANSA, Accessed 5 October 2026. [Online]. Available: https://www.ansa.it/canale_salutebenessere/notizie/sanita/2024/02/26/il-defibrillatore-del-118-arriva-con-il-drone_f33836ad-4a75-4bba-b190-5a87841a99fb.html
- [43] *Droni, nel cremonese il primo servizio di trasporto da una farmacia del territorio*, Pharmacy Scanner, Accessed 5 October 2026. [Online]. Available: <https://pharmacyscanner.it/droni-nel-cremonese-il-primo-servizio-di-transporto-da-una-farmacia-del-territorio/>
- [44] *Boom in Italia dei test per il trasporto via drone di materiale medico*, Supply Chain Italy, Accessed 5 October 2026. [Online]. Available: <https://www.supplychainitaly.it/2022/11/03/boom-in-italia-dei-test-per-il-transporto-via-drone-di-materiale-medico/>
- [45] *Consegna farmaci con droni, in Abruzzo: Il primo volo operativo*, Farmacista33, Accessed 5 October 2026. [Online]. Available: <https://www.farmacista33.it/aggiornamento-professionale/31676/consegna-farmaci-con-droni-in-abruzzo-il-primo-volo-operativo.html>
- [46] *Ospedale San Raffaele sperimenta droni per logistica sanitaria*, TrasportoEuropa, Accessed 5 October 2026. [Online]. Available: <https://www.trasportoeuropa.it/notizie/logistica/ospedale-san-raffaele-sperimenta-droni-per-logistica-sanitaria/>
- [47] *Centro addestramento nazionale SAPR – Croce Rossa Italiana*, Croce Rossa Italiana, Accessed 5 October 2026. [Online]. Available: <https://cri.it/cosa-facciamo/formazione/centroaddestramento-droni/>
- [48] *Droni per trasportare materiale sanitario: Il progetto della Croce Rossa*, Today.it, Accessed 5 October 2026. [Online]. Available: <https://www.today.it/attualita/droni-croce-rossa.html>
- [49] *Mercato UAS e AAM in Italia con un focus sul trasporto sanitario, torna Roma Drone Conference*, Quadricottero News, Accessed 5 October 2026. [Online]. Available: <https://www.quadricottero.com/2022/11/mercato-uas-e-aam-in-italia-con-un.html>
- [50] *Droni per trasferire merci sanitarie, procede il progetto 'Padova Sandbox'*, ANSA, Accessed 5 October 2026. [Online]. Available: https://www.ansa.it/veneto/notizie/2026/05/06/droni-per-trasferire-merci-sanitarie-procede-il-progetto-padova-sandbox_e868e026-ad45-47a8-ae52-71f1a35a627d.html

- [51] *Droni a idrogeno verde per la logistica sanitaria in Veneto*, Rinnovabili.it, Accessed 5 October 2026. [Online]. Available: <https://www.rinnovabili.it/mercato/le-aziende-informano/droni-idrogeno-verde-logistica-sanitaria-veneto/>
- [52] *Al via progetto per usare i droni per il trasporto di medicinali*, ANSA, Accessed 5 October 2026. [Online]. Available: https://www.ansa.it/sito/notizie/cronaca/2023/03/30/al-via-progetto-per-usare-i-droni-per-il-trasporto-di-medicinali_295d5112-acd1-4783-b021-8f36bfe90016.html
- [53] *Testare i droni per il trasporto aereo di materiale biomedicale: Il Living Lab presso l'Ospedale San Raffaele*, Ospedale San Raffaele, Accessed 5 October 2026. [Online]. Available: <https://www.hsr.it/news/2023/agosto/progetto-droni-trasporto-medicine>
- [54] *Droni per trasportare farmaci tra gli ospedali di Varese*, La Prealpina, Accessed 5 October 2026. [Online]. Available: <https://www.prealpina.it/news/cronaca/384462/droni-per-trasportare-farmaci-tra-gli-ospedali-di-varese.html>
- [55] *Droni e logistica sanitaria: Il caso Miulli apre la via ai trasporti medicali del futuro*, Logisticamente, Accessed 5 October 2026. [Online]. Available: <https://www.logisticamente.it/articoli/56333/droni-logistica-sanitaria-ospedale-miulli/>
- [56] *Trasporto sanitario: È boom di droni in Italia*, Sanità Digitale, Accessed 5 October 2026. [Online]. Available: <https://www.sanita-digitale.com/tendenze/trasporto-sanitario-e-boom-di-droni-in-italia/>
- [57] *Trasporto sanitario ad alta tecnologia: Droni cargo collegano Pozzuoli e Ischia in 15 minuti*, Quadricottero News, Accessed 5 October 2026. [Online]. Available: <https://www.quadricottero.com/2025/02/trasporto-sanitario-ad-alta-tecnologia.html>
- [58] *In Abruzzo consegne di medicinali con droni per una sanità sostenibile ed innovativa*, DronEzine, Accessed 5 October 2026. [Online]. Available: <https://www.dronezine.it/456680/in-abruzzo-consegne-di-medicinali-con-droni-per-una-sanita-sostenibile-ed-innovativa/>
- [59] *Droni per il trasporto medicinali e le emergenze, accordo tra ENAC e Croce Rossa Italiana*, Quadricottero News, Accessed 5 October 2026. [Online]. Available: <https://www.quadricottero.com/2022/09/droni-per-il-trasporto-medicinali-e-le.html>
- [60] *Explanatory note to decision 2022/002/R*, European Union Aviation Safety Agency, Accessed 5 October 2026. [Online]. Available: https://www.easa.europa.eu/sites/default/files/dfu/explanatory_note_to_ed_decision_2022-002-r.pdf
- [61] *Drone delivery, Amazon dice stop ai test in Italia. "Non ci sono le condizioni necessarie"*, Pharmacy Scanner, Accessed 5 October 2026. [Online]. Available: <https://pharmacyscanner.it/drone-delivery-amazon-fa-retromarcia-stop-ai-test-in-italia-mancano-le-condizioni-necessarie/>
- [62] *Consegne sanitarie via droni a idrogeno in un progetto per il Parco della Salute di Torino*, Supply Chain Italy, Accessed 5 October 2026. [Online]. Available: <https://www.supplychainitaly.it/2025/04/18/consegne-sanitarie-via-droni-a-idrogeno-in-un-progetto-per-il-parco-della-salute-di-torino>
- [63] *Trasporto sanitario del futuro: I droni accorciano le distanze*, SocialFarma, Accessed 5 October 2026. [Online]. Available: <https://www.socialfarma.it/trasporto-sanitario-del-futuro-i-droni-accorciano-le-distanze/>
- [64] *Droni a idrogeno per materiale sanitario, testato il sistema Padova Sandbox*, Nordest24, Accessed 5 October 2026. [Online]. Available: <https://www.nordest24.it/padova-sandbox-droni-idrogeno-materiale-sanitario>



Extended Market Analysis

A.1. Value-chain actors

Table A.1: Actors involved in drone delivery of medical goods in Italy, grouped by value-chain stage. [†] Based in Piedmont.

Category	Institution	Scope
Drone manufacturers	SkyEU [†]	Autonomous high-speed eVTOL drones for first aid and critical logistics [18]
	FlyingBasket	Cargo drones; used for blood transport Pozzuoli–Ischia [37]
	H2C	Hydrogen drones and hydrogen supply system for Padova Sandbox [38]
	Dronus	“Radon” drone for medical delivery [39]
	Eurolink Systems	“Beluga” medical delivery drone, with Università Cusano [39]
	IDS Ingegneria Dei Sistemi	Defibrillator-carrying drone (WingBeat project) [40]
Technology partners (role not specified)	Mavtech	INDOOR technical partner; aerial surveillance and operational-support products [19]
	FEV Italia [†]	Industrial partner of the DIMEAS feasibility studies; numerical impact analysis
	Leonardo	VTOL drone demonstration flights in Lazio, with Telespazio and D-Flight [41]
	Caltec	Technical partner of SIS 118 for the SEUAM defibrillator drone [42]
	AP 74 UAV	Drone company involved in the Piadena pharmacy link [43]
Payload / container makers	ABzero	Smart Capsule: sensorised, UN3373-certified container for blood, organs and drugs [20]
	Carpitech	“Philotea” temperature- and atmosphere-controlled capsule [44]
	Air Abruzzo	“Flying Box” transport capsule [44]
	Telespazio	TPZ Air 100 drug box used in U-ELCOME [45]

continued on next page

Category	Institution	Scope
Operators	ABzero	Drone operator for San Raffaele trials; handled ENAC authorisation for INDOOR [23], [46]
	Unmanned4you	Operated Aetos blood-transport flights with FlyingBasket [37]
	DL Droni	Piloted the Piadena pharmacy flights [43]
	Telespazio	Runs the U-ELCOME hospital-to-village drug delivery service [16]
	Croce Rossa Italiana	Own drone fleet and certified pilot school; urgent-drug delivery trials [47], [48]
Airspace, infrastructure and energy	ENAV	Air navigation service provider; coordinated Corus-XUAM [49]
	D-Flight	U-space services (ENAV group); Corus-XUAM partner [49]
	DTA	Corus-XUAM partner, Grottaglie airport test site [49]
	ADPM Drones	“Hangar”: docking and recharging station for automated missions [49]
	Gruppo SAVE	Airport group leading Padova Sandbox [38]
	Heron Air	Operator of Padova airport, Sandbox partner [38]
	Milani; DBA Group	Build the hydrogen production and refuelling unit for Padova [50]
	Stesi	Supervision system, ground logistics and spare-parts analytics for Sandbox [51]
	Torino Caselle and Cuneo Levaldigi airports [†]	Proposed hub and support base for hydrogen drone logistics [24]
Research	PIC4SeR, Politecnico di Torino [†]	Autonomous flight technologies for transporting biological material (INDOOR) [25]
	DIMEAS, Politecnico di Torino (Prof. E. Cestino) [†]	Desk-based feasibility studies for emergency drug delivery in Turin: regulatory platform selection and SORA, risk maps and path planning, payload thermal/vibration loads, cost-safety comparison of car, drone and eVTOL, hub siting [9], [10], [11], [12], [13], [14]
	Università di Torino [†]	INDOOR project partner [19]
	Scuola Superiore Sant’Anna	Parent institution of the ABzero spin-off [23]
	Università Cusano	Co-developer of the “Beluga” drone [39]
	San Raffaele technology centre	Developed the FF2020 drone delivery demonstrations [52]
End users	AOU Città della Salute e della Scienza [†]	INDOOR test flight between hospital sites; proposed hydrogen logistics plan; pharmacy department partner of DIMEAS feasibility studies [21], [24]
	Regional Transplant Centre Piemonte–Valle d’Aosta [†]	INDOOR partner for organ and biological transport [19]
	Ospedale San Raffaele	In-campus delivery of drugs and biological samples [53]
	ASST Sette Laghi	Feasibility study for inter-hospital drug transport [54]

continued on next page

Category	Institution	Scope
	Piadena community pharmacies	Pharmacy-to-pharmacy drone link [43]
	ASL 1 Abruzzo	Hospital-only drugs delivered to remote villages [16]
	ASL Napoli 2	Blood transport to Ischia [37]
	Ospedale Miulli	Demonstration flight of life-saving drugs [55]
	ASL Lecce	Planned Gallipoli–Casarano hospital link [56]
	Ospedale di Patti	Planned link to the Aeolian Islands [56]
	Ospedale Pediatrico Bambino Gesù	Partner in VTOL demonstration flights in Lazio [41]
	SIS 118	Emergency service promoting the SEUAM defibrillator drone [42]
Promoters and funders	Fondazione DOT [†]	Promoter of the INDOOR project [19]
	Centro Nazionale Trapianti	Funds the INDOOR research grant at PIC4SeR [22]
	City of Turin [†]	INDOOR partner; funded the LastMed call [19], [26]
	Regione Veneto	Signatory of the Padova Sandbox agreement [38]
	Regione Lazio	Partner in VTOL demonstration flights [41]
	Regione Campania	Institutional support for the Aetos trial [57]
	EU research programmes	H2020 Flying Forward 2020; U-ELCOMe, co-funded by Horizon [46], [58]
	Abruzzo municipalities and Parco Sirente Velino	Host communities of the U-ELCOMe service [16], [45]
Regulators and authorities	ENAC	National civil aviation authority; flight authorisations; protocol with the Red Cross [59]
	EASA	EU rules for drone operations, Regulation (EU) 2019/947 [60]
	Regione Piemonte [†]	Regional health authority; drone trials subject to benefit–risk analysis [27]
Alternatives and exits	Tecnocad / Tecnomill / Bylogix [†]	LastMed autonomous ground rover for last-mile drug delivery [28]
	Amazon Prime Air	Halted drone delivery tests in Italy [61]

A.2. Trials and planned initiatives

Table A.2: Drone delivery of medical goods in Italy: trials and planned initiatives (status as reported by the cited sources; checked 5 October 2026).

Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Piedmont	Turin, Molinette ↔ San Giovanni Bosco feasibility studies	2020–2024	DIMEAS (Politecnico di Torino); AOU Città della Salute e della Scienza (Pharmacy); FEV Italia	Life-saving drugs, blood-derived medication	Approx. 10 km corridor along the Po, approx. 15 min at 17 m/s	Desk studies (5 MSc theses, 1 conference paper); no flights performed	[9], [10], [11], [12], [13], [14]
Piedmont	Turin, INDOOR	Oct 2024	Fondazione DOT; PIC4SeR (Politecnico di Torino); ABzero; AOU Città della Salute e della Scienza	Biological samples (blood) in sensorised capsule	CTO → Molinette, 500 m, automatic navigation over the city and the Po	Single experimental flight	[19], [21]
Piedmont	Turin, hydrogen drone logistics plan	Apr 2025	AOU Città della Salute e della Scienza (Commissioner T. Schael)	Drugs, small medical loads	Torino Caselle hub, Cuneo Levaldigi support, future Parco della Salute	Proposal; no implementation found	[24], [62]
Abruzzo	Avezzano → Collelongo, U-ELCOME	Jul 2025	Telespazio (Leonardo/Thales); ASL 1 Abruzzo; municipalities of Collelongo, Montereale, Rocca di Mezzo	Hospital-only drugs for frail patients	Hospital to village pharmacy, approx. 20 min	Start of operational phase; further routes announced	[16], [17]
Apulia	Grottaglie → Manduria, Corus-XUAM	by 2022	ENAV; D-Flight; DTA	Medical material	Airport to hospital	Test (reported in 2022)	[49]
Apulia	Taranto, SEUAM	Feb 2024	SIS 118; Caltec (Campanian aerospace consortium)	Automated external defibrillator	Urban, simulated cardiac arrest	First test of experimental project	[42]
Apulia	Lecce, ASL Lecce	2025	ASL Lecce	Not specified in source	Gallipoli–Casarano hospitals	Intended	[56]
Apulia	Acquaviva delle Fonti, Ospedale Miulli	Nov 2025	Ospedale Miulli	Life-saving drugs, biomedical material	25 kg hexacopter; 25 km in 25 min	Operational demonstration	[55]
Campania	Pozzuoli → Ischia, Aetos	Feb 2025	FlyingBasket; Unmanned4you; ASL Napoli 2	Blood, biological samples	17 km in 15 min; 4 flights; 5 kg carried	Second trial phase	[37], [63]

continued on next page

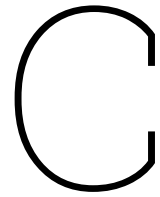
Region	Location / project	Date	Actors	Payload	Route / performance	Reported status	Sources
Lazio	Rome	2022	Leonardo; Telespazio; D-Flight; Ospedale Pediatrico Bambino Gesù; Regione Lazio; Aeroporti di Roma	Not specified in source	Fixed-wing VTOL drones	Two demonstration flights	[41]
Lombardy	Piadena (Cremona)	2021	Two community pharmacies; DL Droni; AP 74 UAV	Drugs and devices between pharmacies	Approx. 6 kg electric drone, approx. 1 kg payload	Announced as continuous service; current status not verified	[43]
Lombardy	Milan, Ospedale San Raffaele, H2020 Flying Forward 2020	Mar 2023	Ospedale San Raffaele; EuroUSC Italia (regulatory consultant); ABzero (operator)	Drugs, biological samples	Within the hospital campus	Demonstration campaign, each flight ENAC-authorised	[46], [53]
Lombardy	Varese, ASST Sette Laghi	2025	ASST Sette Laghi	Drugs, urgent materials	Between two hospitals a few hundred metres apart	Preliminary feasibility study	[54]
Sicily	Trapani, Croce Rossa Italiana	2021	Croce Rossa Italiana	Urgent drugs	Port quay → vessel; 5 min 30 s vs 35 min by sea	Simulation exercise	[48]
Sicily	Patti → Aeolian Islands	2025	Ospedale di Patti	Not specified in source	Patti–Lipari	Under development	[56]
Veneto	Padova, Padova Sandbox	from Mar 2025	Gruppo SAVE; ENAC; Regione Veneto; H2C/H2G; Heron Air; Milani; DBA Group; Stesi	Medical goods, biological samples (hydrogen drones)	Up to 100 km, 4–5 kg payload (stated)	Ground tests of hydrogen system; first flight expected autumn 2026	[38], [64]

B

FEM Iterations for glass phial model

Adding source code to your report/thesis is supported with the package listings. An example can be found below. Files can be added using \lstinputlisting[language=<language>]{<filename>}.

```
1 """
2 ISA_Calculator: import the function, specify the height and it will return a
3 list in the following format: [Temperature, Density, Pressure, Speed_of_Sound].
4 Note that there is no check to see if the maximum altitude is reached.
5 """
6
7 import math
8 g0 = 9.80665
9 R = 287.0
10 layer1 = [0, 288.15, 101325.0]
11 alt = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000]
12 a = [-.0065, 0, .0010, .0028, 0, -.0028, -.0020]
13
14 def atmosphere(h):
15     for i in range(0, len(alt)-1):
16         if h >= alt[i]:
17             layer0 = layer1[:]
18             layer1[0] = min(h, alt[i+1])
19             if a[i] != 0:
20                 layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
21                 layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
22             else:
23                 layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
24     return [layer1[1], layer1[2]/(R*layer1[1]), layer1[2], math.sqrt(1.4*R*layer1[1])]
```



Task Division Example

If a task division is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Table C.1: Distribution of the workload

Task		Student Name(s)
	Summary	
Chapter 1	Introduction	
Chapter 2		
Chapter 3		
Chapter *		
Chapter *	Conclusion	
Editors		
CAD and Figures		
Document Design and Layout		

Acknowledgements - Ringraziamenti

ciao